

Two-Behavior (“Double-Complier”) LATE — 1-Page Skeleton

Setup (exactly as in class).

- Instrument $Z \in \{0, 1\}$ is randomly assigned. We observe two mutually exclusive binary behaviors $D_1, D_2 \in \{0, 1\}$ with $D_1(Z) + D_2(Z) \in \{0, 1\}$ (cannot take both). The realized treatment vector is $(D_1, D_2) = (D_1(Z), D_2(Z))$.
- Potential outcomes: $Y(d_1, d_2)$ and *exclusion* (no direct Z effect): $Y = Y(D_1(Z), D_2(Z))$.
- Random assignment: $(D_1(0), D_1(1), D_2(0), D_2(1), Y(\cdot, \cdot)) \perp Z$.
- (Sometimes given) *Monotonicity*: no defiers *for the behavior of interest*. For “any-treatment” define $T(z) \equiv D_1(z) + D_2(z)$ and monotonicity means $T(1) \geq T(0)$.

Principal strata (two Z -worlds, multiple behaviors).

NT:	$(D_1, D_2) = (0, 0)$ under both z
Always- D_1 :	$(1, 0)$ under both z
Always- D_2 :	$(0, 1)$ under both z
D_1 -compliers:	$(0, 0) \rightarrow (1, 0)$
D_2 -compliers:	$(0, 0) \rightarrow (0, 1)$
(if no mono) D_1 -defiers:	$(1, 0) \rightarrow (0, 0)$, (if no mono) D_2 -defiers: $(0, 1) \rightarrow (0, 0)$

First-stages (net shifts induced by Z).

$$\begin{aligned}\pi^{(1)} &\equiv E[D_1 | Z=1] - E[D_1 | Z=0] = \Pr(D_1\text{-comp}) - \Pr(D_1\text{-def}), \\ \pi^{(2)} &\equiv E[D_2 | Z=1] - E[D_2 | Z=0] = \Pr(D_2\text{-comp}) - \Pr(D_2\text{-def}), \\ \pi^{(T)} &\equiv E[T | Z=1] - E[T | Z=0] = \pi^{(1)} + \pi^{(2)}.\end{aligned}$$

Reduced form for the outcome.

$$\begin{aligned}\delta &\equiv E[Y | Z=1] - E[Y | Z=0] \\ &= \underbrace{E[Y(1, 0) - Y(0, 0) | D_1\text{-comp}] \Pr(D_1\text{-comp})}_{Z \text{ creates } D_1} + \underbrace{E[Y(0, 1) - Y(0, 0) | D_2\text{-comp}] \Pr(D_2\text{-comp})}_{Z \text{ creates } D_2} \\ &\quad - \underbrace{E[Y(1, 0) - Y(0, 0) | D_1\text{-def}] \Pr(D_1\text{-def})}_{\text{if present}} - \underbrace{E[Y(0, 1) - Y(0, 0) | D_2\text{-def}] \Pr(D_2\text{-def})}_{\text{if present}}.\end{aligned}$$

Wald ratios and interpretations. Always state the assumption you use.

$$\begin{aligned}\frac{\delta}{\pi^{(1)}} &= \begin{cases} E[Y(1, 0) - Y(0, 0) | D_1\text{-compliers}], & \text{if } D_1 \text{ monotonicity holds;} \\ \text{signed avg. over } D_1\text{-compliers/defiers}, & \text{without monotonicity.} \end{cases} \\ \frac{\delta}{\pi^{(2)}} &= \begin{cases} E[Y(0, 1) - Y(0, 0) | D_2\text{-compliers}], & \text{if } D_2 \text{ monotonicity holds;} \\ \text{signed avg. over } D_2\text{-compliers/defiers}, & \text{without monotonicity.} \end{cases} \\ \frac{\delta}{\pi^{(T)}} &= \begin{cases} \frac{E[\tau_1 | D_1\text{-comp}] \Pr(D_1\text{-comp}) + E[\tau_2 | D_2\text{-comp}] \Pr(D_2\text{-comp})}{\Pr(D_1\text{-comp}) + \Pr(D_2\text{-comp})}, & \text{mono. in } T, \\ \text{signed mix incl. defiers}, & \text{otherwise,} \end{cases}\end{aligned}$$

where $\tau_1 \equiv Y(1, 0) - Y(0, 0)$ and $\tau_2 \equiv Y(0, 1) - Y(0, 0)$.

“Goofy” reduced forms you can deploy fast.

$$\begin{aligned}\frac{\delta^{X \cdot D_1}}{\pi^{(1)}} &= E[X | D_1\text{-comp}], & \frac{\delta^{Y \cdot D_1}}{\pi^{(1)}} &= E[Y(1, 0) | D_1\text{-comp}], \\ -\frac{\delta^{Y \cdot (1 - D_1 - D_2)}}{\pi^{(1)}} &= E[Y(0, 0) | D_1\text{-comp}] \quad (\text{analogous formulas for } D_2).\end{aligned}$$

Exam checklist (what to do on the page).

1. *Table*: rows = strata (NT, always- D_1 , always- D_2 , D_1 -comp, D_2 -comp, $\pm$ defiers), columns = $Z=0, 1$, fill (D_1, D_2) and $Y(\cdot, \cdot)$.
2. *First stage(s)*: write $\pi^{(1)}, \pi^{(2)}, \pi^{(T)}$; interpret as net shares moved.
3. *Reduced form*: difference the two Z columns; cancel NT/always-takers; keep complier (and defier if any) terms.
4. *Ratio*: pick the ratio the question asks for and state the assumption (mono or not) and the correct interpretation.
5. *Optionally* recover complier means with the goofy products above.

Nested (provider $\rightarrow$ treatment) view you might see. Sometimes Z first changes a setting $H(Z) \in \{0, 1\}$ and treatment is $D(H)$. Then $D_Z = D(H(Z))$, $Y = Y(D(H(Z)))$, and the Wald ratio using D as the first stage identifies the LATE for those who are moved into the setting *and*, because of it, into the treatment (“double compliers”), under random assignment, exclusion, and monotonicity for the final uptake.

Rule of thumb: two counterfactual worlds ($Z=0$ vs $Z=1$); possibly many binary behaviors. Every behavior has a $D_k(0)$ and a $D_k(1)$.